



National Housing Authority (NHA)

82, Segunbagicha, Dhaka-1000, Bangladesh

Draft Terms of Reference (TOR)

for

**“Digitalization of Plot/Flat/Parking/Commercial Space/Shops
Allotment process of National Housing Authority (NHA)”**

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1. Background

Digitization and Automation, as well as the collaborative approach, is one of the most important tools for any institutional management in the age of Information & Communication Technology (ICT). Bangladesh is ensuring an ICT based society where online based management will be more efficient, which will ensure more productivity, transparency, and accountability. One of the significant goals of ICT in Bangladesh is to establish technology-based digital governance (e-Governance).

To resolve these issues and in line with technology driven governance in Bangladesh, the National Housing Authority (NHA) is going to introduce Digitalization of Plot/Flat/Parking/Commercial Space/Shops Allotment process of National Housing Authority (NHA)” to simplify the process of housing allocation.

2. About National Housing Authority (NHA)

National Housing Authority is an autonomous organization under the Ministry of Housing & Public Works of the Government of the Peoples’ Republic of Bangladesh. It was formed in 2001 by virtue of The National Housing Authority Act 2000. Formerly it was known as Housing & Settlement Directorate. National Housing Authority has the mandate to solve the acute housing problem of the urban & rural areas of the country including the capital city of Bangladesh. It’s mission is to ensure safe, sustainable and affordable housing for all strata of people by proper planning on the land available at hand and also to safeguard the property of the Government from illegal possession. NHA’s stakeholders are the people of Bangladesh, contractors & real estate developers, banks & financial institution. National Housing Authority operates all over the country with 10 executive Engineers Offices & 10 Administrative Officers Offices.

3. Scope of work

Teletalk Bangladesh Ltd is required to develop the online allotment system software by following proper & industry standard software development process and software engineering approach step by step. The steps are to be followed for development and implementation of the overall system can be divided into below phases:

- System Requirement Analysis
- System Design
- Development
- Integration
- Testing
- Hosting
- User Acceptance Test (UAT)

- Deployment and Implementation
- Training and Knowledge Transfer
- OAT (Operation Acceptance Testing) and Commissioning
- Maintenance and Support Service

4. Proposed System Requirements

4.1 Digitalization of Plot/Flat/Parking/Commercial Space/Shops Allotment process of National Housing Authority (NHA)”

Existing Process:

At present, Service seekers apply to NHA for flat/plot allotment whenever a circular is published under a project. These applications are taken manually through selected banks. The aspirants have to visit bank in person to buy prospectus and application. The applications are submitted at banks with security money payment. Those applications are sent to NHA. NHA then perform scrutiny of applications, sort the applications, create valid applicant list and lottery is performed at NHA offices in front of the applicants in traditional way. The lottery winners are then awarded allotment paper by NHA.

Proposed System:

Teletalk Bangladesh Ltd is required to transform this service into an online system where aspirants can buy prospectus by paying fees, apply for desired plot/flat/parking/commercial space/shops by filling out necessary information for applying. A user profile will be created after NID verification for an applicant. The system will show available project for allotment. The applicant will choose the desired project and buy prospectus online. After buying prospectus, if interested, the applicant will fill out the necessary information online to submit application. Necessary fields for applying for allotment is given below (but not limited to):

1. Name (English and Bengali)
2. Gender
3. Father's Name
4. Mother's Name
5. Husband/Wife's Name
6. NID No
7. Birth Date
8. Flat Size (in sqft)/ Plot Size (in Katha) (selection type: Dropdown/radio buttons etc)
9. Halafnama (Upload)
10. Pay Order/Bank Draft for security (Upload field/Pay Online)
11. Quota selection (if implied in future)
12. Permanent Address

13. Present Address

14. Occupation

15. Other application fields (subject to detail software requirement analysis)

A sample application (Annexure-1) is attached herewith for requirement analysis for required application fields. After filling out all the information, the applicant will successfully submit the application and the application will be marked by a unique application number which will later be used in the lottery process. The submitted application will be received at NHA's end in the system. The application field's can vary from project to project depending on the nature of the project. For example, In flat project's application Flat size in sqft will be available for selection and in plot project's application, plot size in katha will be available for selection. These fields will be configured from NHA's admin panel/admin user. A deadline will be set for submitting application after which the system will be closed for new application. Option for extending the deadline will be available, if needed.

After the deadline, NHA's system users will perform scrutiny on the submitted applications. The scrutiny committee users will be able to discard invalid applications based on required justifications. After the scrutiny of applications, valid application list will be available in the system for all users and the users will be notified via email & phone no. If any project has approved quota in the prospectus, applicant will be able to apply under quota from selection. After scrutiny by the committee, valid applicant list will be generated under certain quota and lottery will be performed separately under different quota category.

After the scrutiny of valid and invalid applications, the authority will decide to perform online lottery on a certain day & applicants will be notified via this system of the date and time of the lottery. Teletalk Bangladesh Ltd is required to develop an online lottery system through which winners of the lottery will be picked automatically without any human interference. The online lottery system will choose an applicant number and a flat no/plot no/parking etc both automatically using efficient and industry proven algorithm and process. The lottery algorithm must be designed in such a manner that it will have absolute transparency and subject to any public auditing. Recognized organization/institutes will test the algorithm's efficiency, transparency and characteristics of whether the algorithm is using any common pattern or not. Only after passing the test, the online lottery will take place in the overall system.

After automatically picking each winner applicant and corresponding housing allocation, total lottery will mark an end and generate winner list with all details right after the lottery process. The applicants will be notified through the system about the lottery result and their lottery status will be updated in their applicant profile in the system.

The online lottery result will get final approval from the authority and the winner applicants will be provided allotment notice and notification through the system.

The system will store all the information of each project's applicants, valid/invalid list, lottery winner list for NHA users.

Payment System:

Teletalk Bangladesh Ltd will have to develop a secure payment gateway for the payment of prospectus and security money. Major Cards Processors, Banks and Mobile Wallets/m-Banking must be available so that citizen can pay with their DEBIT/CREDIT/VISA card or proprietary card (like DBBL Nexus, qCash, etc.) and other way of online payments (as payment through POSs) through this channel. Provision of payment by Mobile wallet (like bKash, DBBL, Mobile, etc) must be present on the Online Payment Gateway. Incorporate bank to bank transfer as soon as the scope is available. Need transaction log and audit trails for the payment system, protect sensitive financial information, and ensure compliance with laws and regulations. For security money offline payment, include provision for uploading pay order/bank draft in the system. Provide provisions for bank reconciliation for security money if possible.

Online Lottery System:

The online lottery process will be conducted using the lottery algorithm developed by Teletalk Bangladesh Ltd. An online lottery will be conducted among the valid applicants. Through this lottery, an applicant's serial number will be selected sequentially and neutrally, and a flat/plot/parking/commercial space number will then be impartially assigned against that selected serial number. The lottery will select as many winning applicants' serial numbers as there are flats/plots/parking spaces/commercial spaces available for allotment. At the end of the process, the lottery results will be generated automatically and displayed on the system. The lottery algorithm and the entire process will be thoroughly tested and certified by BUET, the University of Dhaka, and the Bangladesh Computer Council.



Figure 1: Online Lottery conceptual overview

Dashboard:

Teletalk Bangladesh Ltd will develop interactive dashboard for service recipients as well as for the officials of NHA for this system. There can be different types of dashboard templates based on user layers and types and systems. From Admin/Officials dashboard NHA users will be able to view applicants/no of applicants, manage & perform all necessary process like scrutiny of applications, perform lottery and issue allotment paper but not limited to these for the above-

mentioned proposed system. Dashboard must be designed in such a way that it can be used to monitor performance, track progress, and make decisions and generate dynamic report.

Reporting System:

Teletalk Bangladesh Ltd will implement dynamic reporting system in the system so that only NHA officials can generate all types of reports for this system. Necessary fields and format of the report will be approved by NHA. All reports must be in the downloadable and printable format (pdf/excel etc). The reporting system must provide flexibility to the official user, so that they can choose fields for the format of the report from multiple dropdowns, checkboxes etc. Typical reporting outcome from the system may include (but not limited to): Valid/invalid applicant list of a circular under a project, lottery winner list with applicant number and flat/plot/parking number and other details of a circular under a project, sent sms notification report

The possible process flow for transforming the traditional lottery system into a automated and digitalized lottery is given below:

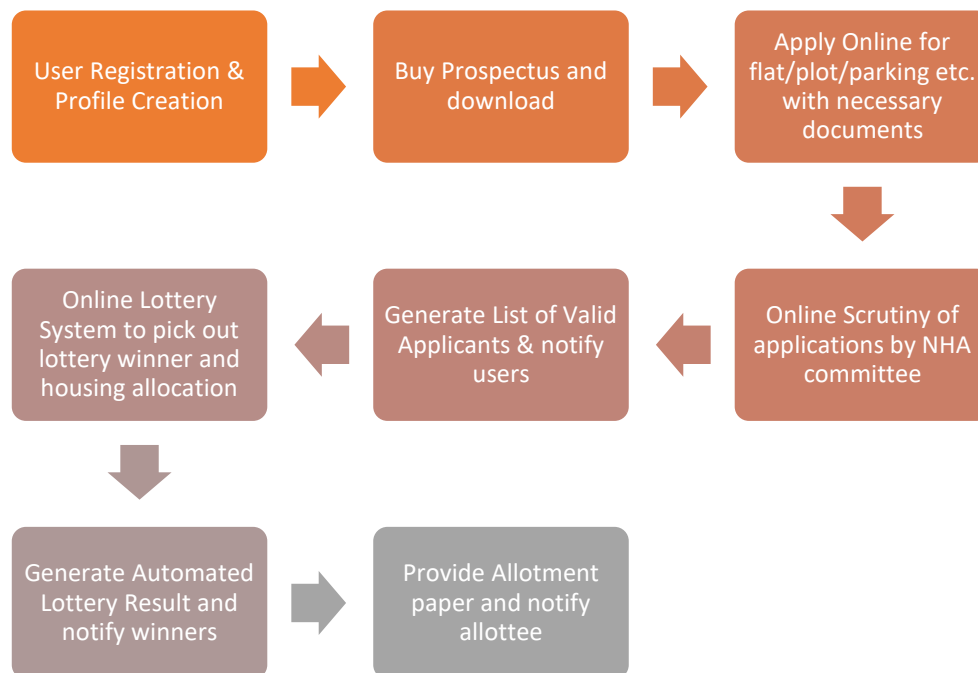


Figure 2: Proposed Process flow overview for Digitalization of Plot/Flat/Parking/Commercial Space/Shops Allotment process of National Housing Authority (NHA)

4.2 Common Services Configuration

Common services configuration refers to the setup and configuration of services that are commonly used in an organization or network. They are often designed to be reusable, modular, and easily configurable, allowing them to be easily integrated into different systems and applications. By using common services, organizations can reduce the costs and complexity associated with developing and maintaining separate, standalone services for each application or system.

4.2.1 User Management

User management is the process of creating, modifying, and maintaining user accounts for an organization's system, software, or website. This includes tasks such as creating new accounts, assigning roles and permissions, resetting passwords, and deactivating or deleting accounts. User management is an important aspect of ensuring the security and access control of an organization's systems and data.

User Account Management

User account management should have the following options:

- Insert, update, block, delete users -Users can register in the system.
- System admin can assign users to their designations with designated roles.
- System admin can assign users to their designations with designated roles.
- System admin can release user from his/her designation.

User Role Management

Insert, update user roles. Assign different types of roles to different kinds of users.

User Account Verification

There will have option to verify users based on email and/or SMS.

User Account Recovery

Easy implementation of a "Forgot your password?" feature in this Platform can choose between two methods: using email using SMS based secret code.

Change Password

Any user can change his/her password any time to address security challenges.

Profile Management

System will have provision to add and update user account information such as the password, email, phone number, photo, signature etc. Users will be able to login and view submitted application details from the system and view application status.

4.2.2 Access Control Management

Access control management is the process of controlling access to resources or systems within an organization. This includes identifying and authenticating users, granting or denying access to specific resources, and monitoring and logging access attempts. Access control management helps to ensure the security and integrity of an organization's systems and data by preventing unauthorized access or changes.

Multi-Layer Access Control Mechanism

The system will have system access control panel. Every system user will be assigned to one or multiple roles. Every role should be assigned to a set of actions or activities. The system admin will have provision to assign roles to users and to assign set of actions/activities to roles. There

should have provision to assign layer, section, department, office-based access management system

4.2.3 Notification Management

Notification Medium

There are two mediums of notifications- SMS, e- Email, SMS: Sending SMS to service consumer and provider's mobile phone in real time is an important concern to process and deliver services efficiently. Email: Proper and timely notification is a vital concern at required point of each service's life cycle. Email notification is very usual among 3 types of notifications like Email, SMS and System notification. There should have template with specific format for these notifications. The systems can access Queue manager and Enterprise Bus for these notifications.

Notification Types

System generated notification is another important concern and should be ensured along with Email/SMS notification in service life cycle Notification can be 3 types configurable, call to action, alert und medium can be via e-Mail, SMS and application systems.

4.3 Integration Requirements

Teletalk Bangladesh Ltd will integrate the following External System but not limited to:

SL	External System	Integration Scope
1.	NID	Integration with NID server of Election Commission of Bangladesh/concerned authority. Necessary agreement is already signed between NHA & Election Commission for API.
2.	Payment gateway	Payment Gateway integration with all kinds of cards, MFS, bank to bank transfer facilities for payment of installments, service fee.

5. Functional Requirements

This online system will ensure below functional requirements so that, the users of this system and monitoring authority get the desired service expected on each part. Apart from the business requirements, the software must fulfil other basic functional requirements of a successful and a full proof system. Below are the functional requirements, that need to be addressed and implemented properly in this system to ensure smooth operation and sustainability.

5.1 Security and Privacy Requirements

The authentication and authorization of the system need to be robust enough to ensure highest level of security. The system prevents all standard web vulnerabilities and provides industry standard security measurements. With a very strong Enterprise system's regular security audit and

careful implementation of various measures least but not limited to the following must be taken to prevent any kind of security breach.

5.1.1 Application Security

5.1.2 Data Security

6. Non-Functional Requirements

The system must ensure all industry standard non-functional requirements of a public service oriented software. The online application should easily be accessible online with user friendly and highly responsive UI and all types of browser support. To ensure sizing, performance and scalability the system shall be able to handle simultaneous 500/1000 users with minimum load and response time. It should be operable at very low bandwidth and system's data growth estimation need to be considered while designing the system. For system audit purpose, this online system will maintain an audit trail of any changes or updates made in any information that are considered as vital and should maintain the audit log with information. The system should support multilingual option for language support.

6.1 Tools and Technologies to be used

Understanding the details scope of this project, Teletalk Bangladesh Ltd is requested to submit a detail "Tools & Technology" plan in their technical proposal following the table format mentioned below

Issues/Phases/Purpose	Used Technology/ Tools	Justification for use
Project Management		
Version Control		
System Requirement Analysis		
System Design		
Development (Client end)		
Development (Server end)		
API/Web services		
Testing		
Documentation		
QA		
Helpdesk/Support		

7. Expected Outputs/Deliverables

Considering the scope of works the consultant is requested to deliver as part of this service / assignment mentioned below, (deliverables proof will be fixup as per client's requirement):

Deliverable No.	Deliverables
D1.	Inception report with service/assignment management plan (including finalized methodology, detail work plan, Gantt Chart etc).
D2.	Detailed System Requirement Specification (SRS) in standard format consisting External Interface Requirements, System Features, Nonfunctional Requirements and Other Requirements (if any) based on assignment objectives, scope of works & proper documented customers' requirements for all the systems stated in this ToR.
D3.	Detail System Design Documents (HLD and LLD as Software Design Descriptions [SDD] IEEE standards).
D4.	Development & Delivery of a web based solution for Digitalization of Plot/Flat/Parking/Commercial Space/Shops Allotment process of National Housing Authority (NHA) as per requirement stated in ToR and agreed on SRS.
D5.	Test plan with its testing acceptance (consultant, user and third-party organization nominated by client like Bangladesh Computer Council/BUET or others) report based on the detailed required documents. Fix any issues or bugs in this section.
D6.	Complete System Installation & Deployment on Live Server.
D7.	Final version of the software & system along with the properly documented source code, database & other required credentials, user manual (text and/or video).
D8.	Training and Technology Transfer with Source code and training manual
D9.	Service/Assignment completion report with upcoming challenges

8. Project Management

8.1 Project Development Timeline

Teletalk Bangladesh Ltd will propose the development timeline of this system.

8.2 Project Implementation Timeline/Schedule

Teletalk Bangladesh Ltd will submit project implementation timeline/schedule in below format to NHA.

Development Phase (____ Months)			
SL	Activity	Duration	Total Weeks
1	Inception Report		
2	System Requirement Specification		
3	System Design		
4	Software Development, Beta Version Release, Testing& Final version of Software Release		
5	UAT and Training		
6	OAT and Commissioning		
7	Training & Knowledge Transfer		

9. Copyright & Ownership

National Housing Authority (NHA) shall be entitled to all proprietary rights including but not limited to patents, copyrights and trademarks.

NHA will hold the copyright claim and be the owner of all kinds of source code including code documentation and other approved documents (all versions trail, products, developed applications, documents and all kinds of deliverables which bear a direct relation to or is made in consequence of the services provided by Teletalk Bangladesh Ltd under this scope of this TOR.

10. Conclusion

National Housing Authority (NHA) expect that Teletalk Bangladesh Ltd will complete this project in time and help to serve the citizen's services and the ultimate expectation is to automate business functions facilitating improved services to its clients through online. Teletalk Bangladesh Ltd has to ensure that all technical difficulties should address in professional and effective manner. NHA is trying to ensure faster, systematic and much higher consumer service delivery by introducing a faster, robust, secure, tamper-proof and efficient Plot/Flat related services technology framework. Future scope of managing after services of allottees after the allotment can be extended to information regarding payment/installment collection of allocated plot/flat/others through this system and Teletalk Bangladesh Ltd can engage with NHA by proposing business continuity plan.